



Figure 11: Wireshark interface

4. Allow your computer to accept the X Server connection. For example, if the NST Linux IP address is 192.168.0.105 and the client is 192.168.0.104, run the `xhost` command from the customer side to include the NST Linux server in the rundown of permitted hosts to make the connections.

```
$ xhost +192.168.0.105
```

Once you have done this, you can attempt to dispatch the X Window application from the NST WUI. For instance, we can attempt dispatching the Wireshark application from the NST WUI. Go to the menu and then perform the following steps: `X > Network Applications > Wireshark (Packet Capture)`.

At this point, Wireshark will show up.

The status (on probe) at the header reveals to us that it really begins from the server, yet is rendered on the client side. On the off chance that you are running the Microsoft Windows client, you can do the same on it, if you also run Cygwin/X on your Windows client. Reboot or shut down the server.

NST WUI also allows the server administrator to restart or shut down the server from the Web. If a server reboot is required, go to `System > Control Management > Reboot`.

Using NST in the wild

Let's now look at the different uses of NST in a wide variety of network environments.

Basic use case 1: This is a simple configuration for NST. A small computer like a notebook is attached directly to a broadband cable network. This configuration is helpful for checking and exploring the Intrusion Detection System (IDS).

Basic use case 2: There is another basic simple configuration, which involves a notebook computer running NST behind a router, switch, firewall or wireless device that is attached to a broadband cable network. This set-up is valuable for investigating the NST Linux operating system and its abilities.

Mobile wireless monitoring: This involves a notebook

computer running NST to monitor 802.11 wireless networks. This plan is alluring for running the Kismet 3 remote network sniffers.

Small business configuration: This is a commonplace NST set-up and provides network security observation inside a private business network environment that is joined to the public Internet.

Enterprise configuration: This type of configuration gathers all the information from the network. A corporate enterprise network helps connect computers and related devices across departments and workgroups. For securing data or information, NST can be integrated in the corporate enterprise network environment.

Using VPNs with NST

VPN tunnels have for some time been used to secure and ensure the integrity of information over untrusted systems like the Internet. By using distinctive VPN connection types with NST, what you access on the Web and anything sent from one system to another is encrypted and directed through the VPN. So, data sent to the network can't be read by anyone except the VPN provider. When a connection is not encrypted, an attacker could perform a man-in-the-middle exploit (MITM), whereby the attacker can see all the information that is not encrypted being sent to other networks—including user names and passwords.

Let's now look at an instance of using VPN with Point to Point Protocol (PPP) over SSH.

VPN: PPP tunnelled over SSH

Probably the most common VPN solutions are SSH-Tunnel, PPP, PPTP and OpenVPN. I personally think that OpenVPN is the best option since it's strong and secure. The Secure Shell (SSH) can likewise be used to make an encoded tunnel between two PCs. PPP over SSH is a fast and speedy VPN solution. You can run a PPP connection over an SSH connection to make a simple, encrypted VPN. With PPP and SSH tools, you can create your own VPN within just a few minutes. Before I get into the details, shown below is a sample of PPP over SSH in a network.

Here, I will use a Windows machine to demonstrate the PPP tunnel over SSH in a VPN network. The steps are as follows.

1. Set up the VPN. The commands given below should run on a remote NST probe.

```
Script:"vpn-ppps" Is Run On Remote NST probe:"192.168.1.51"
SSH Configuration File:/root/.ssh/config
HOST nstprobe
HostName=70.22.33.10
Port=20022
VPN PPP SSH Script: vpn-ppps
Vpn-ppsh -r nstprobe -s 172.18.2.31 -c 172.18.2.32 -rt -sn \
172.18.2.0/24 -cn 192.168.1.0/24 -nt -v
```

2. Once the VPN is set up, you can connect the two NST